Scaling Book Exercises: Section 8

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Chapter: Serving LLaMA 3-70B on TPUs
Source scan: `../handwritten/section_08_handwritten.pdf`
Reference: `https://jax-ml.github.io/scaling-book/applied-inference/`

TPU v5e constants. Each chip has 16 GB of HBM, bfloat16 peak compute

$$C = 1.97 \times 10^{14} \text{ FLOPs/s},$$

and HBM bandwidth

$$W_{\text{HBM}} = 8.2 \times 10^{11} \text{ bytes/s}.$$

Question 1: LLaMA 3-405B lower bounds

Let $P = 405 \times 10^9$ be the parameter count. A transformer forward pass uses approximately two FLOPs per parameter per token, so

$$\boxed{\text{FLOPs/token} \approx 2P = 810 \times 10^9}.$$

On N TPU v5e chips, the ideal compute-bound latency is

$$\begin{aligned} T_{\text{math}} &= \frac{2P}{NC} \\ &= \frac{810 \times 10^9}{N(1.97 \times 10^{14})} \\ &= \boxed{\frac{4.11}{N} \text{ ms/token}}. \end{aligned}$$

If parameter loading is the bottleneck, assume bfloat16 parameters. Their total size is $2P = 810$ GB, so the ideal HBM-bound latency is

$$\begin{aligned} T_{\text{parameters}} &= \frac{2P \text{ bytes}}{NW_{\text{HBM}}} \\ &= \frac{810 \times 10^9}{N(8.2 \times 10^{11})} \\ &= \boxed{\frac{988}{N} \text{ ms/token}}. \end{aligned}$$

Both are optimistic lower bounds: they assume perfect parallel scaling and ignore collective latency and other runtime overheads.

Question 2: Serving LLaMA 3-8B at batch size 240

Assume int8 weights, int8 KV caches, and an 8,192-token context. The relevant LLaMA 3-8B dimensions are

$$L = 32, \qquad K = 8, \qquad H = 128, \qquad V = 128{,}256.$$

(a) Model parameters

At one byte per int8 parameter,

$$\boxed{M_{\text{parameters}} = 8 \times 10^9 \text{ bytes} = 8 \text{ GB}}.$$

(b) KV caches

Each token stores one key and one value for every layer and KV head:

$$M_{\text{KV/token}} = 2LKH = 2(32)(8)(128) = 65{,}536 \text{ bytes}.$$

Thus one 8,192-token sequence requires

$$65{,}536(8{,}192) = 536{,}870{,}912 \text{ bytes} \approx 0.537 \text{ GB}.$$

At batch size $B = 240$,

$$\boxed{M_{\text{KV}} \approx 128.85 \text{ GB}}.$$

(c) Peak working activations

The largest unsharded working activation is approximately the logits tensor $[B, V]$. Stored in bfloat16, it occupies

$$M_{\text{activations}} \approx 2BV = 2(240)(128{,}256) = 61{,}562{,}880 \text{ bytes},$$

or

$$\boxed{M_{\text{activations}} \approx 61.6 \text{ MB}}.$$

This is small compared with the KV cache.

Smallest topology

The total is approximately

$$8 + 128.85 + 0.062 = 136.91 \text{ GB}.$$

The raw capacity lower bound is therefore

$$\left\lceil \frac{136.91}{16} \right\rceil = 9 \text{ chips}.$$

The next supported TPU v5e topology is a 4×4 slice, so the smallest usable topology is

$$\boxed{4 \times 4 = 16 \text{ chips}}.$$

Question 3: Serving LLaMA 3-405B

Assume int8 weights, bfloat16 FLOPs, and a 128-Ki-token ($S = 131{,}072$) context. LLaMA 3.1-405B has

$$P = 405 \times 10^9, \qquad L = 126, \qquad K = 8, \qquad H = 128, \qquad F = 53{,}248.$$

Memory

The int8 model requires 405 GB. The KV cache per token is

$$2LKH = 2(126)(8)(128) = 258{,}048 \text{ bytes.}$$

One full-length sequence therefore requires

$$258{,}048(131{,}072) = 33.82 \times 10^9 \text{ bytes} = 33.82 \text{ GB.}$$

For batch size B and N chips, the memory constraint is

$$405 + 33.82B \le 16N.$$

Even $B = 1$ requires at least

$$\left\lceil \frac{405 + 33.82}{16} \right\rceil = 28 \text{ chips,}$$

so the minimum supported topology is $4 \times 8 = 32$ chips.

Latency model

The ideal generation-step model is

$$T_{\text{step}} = T_{\text{KV}} + \max(T_{\text{parameters}}, T_{\text{math}})\,.$$

In milliseconds,

$$\begin{aligned}
T_{\text{parameters}} &= \frac{405 \times 10^9}{N(8.2 \times 10^{11})}(10^3) = \frac{493.9}{N},\\
T_{\text{KV}} &= \frac{B(33.82 \times 10^9)}{N(8.2 \times 10^{11})}(10^3) = \frac{41.25B}{N},\\
T_{\text{math}} &= \frac{2B(405 \times 10^9)}{N(1.97 \times 10^{14})}(10^3) = \frac{4.11B}{N}.
\end{aligned}$$

Compute overtakes parameter loading only near

$$4.11B = 493.9 \quad \Longrightarrow \quad B \approx 120.$$

All feasible points below use $B < 120$, so

$$T_{\text{step}} \approx \frac{493.9 + 41.25B}{N} \text{ ms.}$$

The largest batch satisfying both memory and the 15 ms latency limit is summarized below.

N	Topology	B_{mem}	B_{latency}	Chosen B	Step time
32	4×8	3	none	—	> 15 ms
64	8×8	18	11	11	14.81 ms
128	8×16	48	34	34	14.82 ms
256	16×16	109	81	81	14.98 ms

Ignoring ICI communication, the corresponding aggregate throughputs are approximately 743, 2,295, and 5,407 tokens/s. The naive roofline would therefore choose 256 chips at batch size 81.

ICI roofline

For ordinary decode tensor parallelism, the approximate useful sharding limit is

$$p_{\max} \approx \frac{F}{8B}.$$

At $N = 256$ and $B = 81$,

$$p_{\max} \approx \frac{53{,}248}{8(81)} \approx 82 < 256,$$

so that point is ICI-bound and its naive throughput estimate is not achievable. At $N = 128$ and $B = 34$,

$$p_{\max} \approx \frac{53{,}248}{8(34)} \approx 196 > 128.$$

Thus the simpler configuration that remains below this approximate ICI roofline is

$$\boxed{N = 128\ (8 \times 16), \quad B = 34, \quad T_{\text{step}} \approx 14.82 \text{ ms}},$$

with aggregate throughput

$$\boxed{\frac{34}{0.01482} \approx 2{,}295 \text{ tokens/s}}.$$

A 256-chip deployment could instead run multiple smaller replicas; the single-replica $N = 256, B = 81$ calculation should not be treated as a valid pure tensor-parallel operating point.

Theoretical minimum step time

The most optimistic latency occurs at $N = 256, B = 1$:

$$\begin{aligned} T_{\text{parameters}} &\approx 1.929 \text{ ms}, \\ T_{\text{KV}} &\approx 0.161 \text{ ms}, \\ T_{\text{math}} &\approx 0.016 \text{ ms}. \end{aligned}$$

Therefore

$$T_{\min} = 0.161 + \max(1.929, 0.016) = \boxed{2.09 \text{ ms/token}}.$$

This is an ideal bandwidth lower bound; real ICI and fixed overheads make the observed latency larger.